



Kaempferol inhibits atherosclerotic plaque development via dual-targeting of p53-p21-p16 senescence pathway and Nrf2/HO-1/NQO1 antioxidant mechanism: Insights from combined in vivo and in vitro research

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ABSTRACT

Background: Atherosclerosis, a leading cause of cardiovascular disease, is driven by aging-related endothelial cell senescence, oxidative stress, and inflammation. Kaempferol, a dietary flavonoid, exerts potent antioxidant and anti-senescence effects on the vasculature.

Objective: We investigated whether kaempferol mitigates atherosclerosis through dual suppression of vascular oxidative damage and endothelial senescence.

Methods: Male ApoE^{-/-} mice on a high-fat diet were treated with kaempferol or vehicle. Atherosclerotic lesions were quantified in the aorta (Oil Red O staining), and endothelial function was assessed by vasodilation assays. Aortic tissues were analyzed for oxidative stress markers (ROS, MDA, SOD) and inflammatory cytokines (TNF- α , IL-6). Endothelial senescence in the vasculature was evaluated by γ -H2A.X staining and expression of p53, p16^{INK4a} and p21. Separately, human endothelial cells were exposed to H₂O₂ to induce oxidative stress and senescence, with or without kaempferol; ROS levels and markers of cellular senescence (SA- β -gal, p53, p16^{INK4a}, p21) were then measured.

Results: Kaempferol-treated mice exhibited markedly smaller aortic plaque areas than controls. Endothelium-dependent vasorelaxation improved, and oxidative stress in the aorta was attenuated (reduced ROS/MDA, increased SOD activity). Pro-inflammatory TNF- α and IL-6 levels were suppressed, while antioxidant Nrf2 and HO-1 expression was elevated. In vitro, kaempferol significantly reduced H₂O₂-induced endothelial senescence: treated cells showed fewer SA- β -gal-positive cells and decreased expression of p16^{INK4a} and p21, indicating reduced cellular aging.

Conclusion: Kaempferol significantly ameliorates experimental atherosclerosis via simultaneous antioxidative and anti-senescent actions, highlighting its potential as a therapeutic agent for vascular disease.

1. Introduction

Atherosclerosis (AS) remains the leading pathological basis of cardiovascular diseases, results from the dynamic interplay of dyslipidemia, oxidative stress imbalance, and vascular cell senescence [1]. Despite the significant improvement in clinical outcomes due to statins lowering low-density lipoprotein cholesterol (LDL-C), 30–40 % of patients continue to face residual cardiovascular risk, experiencing plaque progression or vulnerability even with optimal lipid-lowering therapy [2,3]. Statins' ability to stabilize advanced plaques is limited, residual cardiovascular risk remains substantial [4,5], and long-term use can

worsen insulin resistance and increase type 2 diabetes incidence in non-diabetic individuals [6,7]. These limitations underscore the urgent need for therapeutic strategies that address the residual risk and plaque instability unresolved by lipid-lowering alone.

Emerging evidence indicates that oxidative stress and cellular senescence accelerate plaque destabilization through an "oxidation-inflammation-senescence" vicious cycle [8–10]. Disrupting this cycle, while simultaneously correcting systemic metabolic imbalances, represents a promising strategy for comprehensive AS intervention. Consequently, natural flavonoids-renowned for their multi-target regulatory properties-have attracted considerable attention as potential

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therapeutic agents in AS [11,12]. Kaempferol (KA), a bioactive flavonol ubiquitous in various medicinal plants, exerts anti-inflammatory and antioxidant effects in vascular contexts [13,14](Fig. 1A). For example, KA inhibits CD36-mediated mitochondrial ROS production by blocking Piezo1 channels and Ca^{2+} influx, thereby modulating MAPK/NF- κ B and Nrf2/HO-1 pathways to suppress foam cell formation [15]. It also alleviates AS through GPER-mediated activation of the PI3K/AKT/Nrf2 pathway [16], promotes M1-to-M2 macrophage repolarization for anti-inflammatory effects [17], and regulates the OPN-CD44 axis to mitigate inflammation [18]. KA has demonstrated endothelial-protective and anti-atherogenic effects: alleviating endothelial injury via the circNOL12/miR-6873-3p/FRS2 axis [19], attenuating ox-LDL-induced apoptosis through miR-26a-5p-mediated inhibition of the TLR4/NF- κ B pathway [20], and reducing lipid accumulation in macrophages by downregulating CD36 while upregulating SR-BI and ABC transporters A1/G1 [21]. However, despite these advances, no study to date has examined whether KA can counteract vascular cell senescence—an emerging and critical driver of plaque instability that remains largely overlooked.

Here, we reveal a novel “metabolic remodeling-redox homeostasis-senescence microenvironment” triple regulatory axis by which KA attenuates AS progression. Using ApoE^{-/-} mouse models, multi-omics profiling, and molecular docking, we provide the first evidence that in AS KA directly targets both Nrf2 and p53, enabling synergistic antioxidant and anti-senescence effects. This dual-target mechanism transcends prior single-pathway studies, offering a mechanistic basis for restoring vascular microenvironment homeostasis through epigenetic reprogramming. Our findings not only identify KA as a candidate for next-generation multi-target anti-AS therapies but also propose a conceptual framework for tackling the residual risk left unresolved.

2. Materials and methods

2.1. Animal experiments

Male C57BL/6 J (wild-type) and ApoE^{-/-} mice (6 weeks old, weighing 18–22 g) were obtained from Jiangsu GemPharmatech Co., Ltd. The mice were kept in an SPF-grade animal facility with controlled environmental conditions (temperature at $22 \pm 1^\circ\text{C}$, humidity at $55 \pm 5\%$, and a 12-h light/dark cycle) with unrestricted access to food and water. The Experimental Animal Ethics Committee of Nanjing University of Chinese Medicine approved all experimental protocols (Approval No. ACU231202, December 18, 2023). ApoE^{-/-} mice were randomly divided into four groups ($n = 12$ each) using computer-generated randomization: Model group (HFD + 0.5 % sodium carboxymethyl cellulose [CMC—Na] solvent control), KA low-dose group (HFD + 10 mg/kg/day KA [MCE, purity >98 %, CAS 520–18–3]), KA high-dose group (HFD + 20 mg/kg/day KA), and the Atorvastatin group (HFD + 10 mg/kg/day atorvastatin [Pfizer, USA]). Age-matched C57BL/6 J mice fed a standard diet served as the control group ($n = 12$).

The HFD (D12108C, Research Diets) contained 40 % fat-derived calories and 1.25 % cholesterol. Drug administration: involved suspending the drugs in a 0.5 % CMC-Na solution and delivering them orally via gavage (200 μL) at a consistent daily time, with the control group receiving an equivalent volume of saline. Following a 1-week acclimatization period, interventions were carried out for 12 weeks. Body weight was measured weekly using a precision electronic balance (± 0.1 g). At the conclusion of the 12-week period, the mice were fasted for 12 h and anesthetized by intraperitoneal injection of 2.5 % tribromoethanol (Avertin, 200 mg/kg). Blood samples were obtained from the retro-orbital venous plexus, centrifuged at 3000g for 10 min at 4°C to separate serum, and then stored at -80°C . The left ventricle was cannulated and perfused with pre-cooled PBS (20 mL, 4°C) to remove circulating blood. The aortic root was dissected, embedded in OCT, and stored at -80°C for subsequent frozen sectioning. Aortas were rapidly frozen in liquid nitrogen and stored at -80°C for later western blot

analysis. Liver tissues were fixed in 4 % paraformaldehyde for 24 h before being paraffin-embedded for Hematoxylin and Eosin (H&E) Staining.

2.2. Histological staining

2.2.1. H&E staining

Aortic root frozen sections were air-dried for 1 h at room temperature, stained with hematoxylin (Servicebio, Wuhan, China) for 10 min, rinsed thrice with tap water, and then stained with eosin (Servicebio, Wuhan, China) for 30 s. Sections were washed thrice with tap water, dehydrated using a graded ethanol series (75 %, 95 %, 100 %), cleared in xylene for 5 min, and mounted under coverslips with neutral balsam (Sinopharm Chemical Reagent Co., Ltd., Shanghai, China). Images were obtained with an optical microscope.

2.2.2. Masson's trichrome staining

Aortic root frozen sections were air-dried for 1 h at room temperature, stained with Weigert's iron hematoxylin (Servicebio, Wuhan, China) for 10 min, differentiated in 1 % hydrochloric acid ethanol for 15 s, and rinsed three times with tap water. Sections were stained using Masson's blue solution for 5 min, followed by three washes with tap water. They were then treated with Ponceau S red solution for 10 min, rinsed with phosphomolybdic acid solution, and finally stained with aniline blue solution for 1 min. All solutions were sourced from Servicebio, Wuhan, China. Sections were dehydrated using a graded ethanol series, cleared in xylene for 5 min, and mounted under coverslips with neutral balsam. Images were obtained with an optical microscope.

2.2.3. Oil red O staining

Frozen aortic sinus sections were brought to room temperature, fixed with tissue fixative (Servicebio, Wuhan, China) for 15 min, rinsed with tap water, and then air-dried. Sections were stained for 10 min in the dark using an Oil Red O working solution, which was prepared by combining a saturated Oil Red O stock solution with distilled water in a 6:4 ratio, thoroughly mixed, and filtered after overnight storage at 4°C , followed by a second filtration after 24 h at 4°C . After staining, sections underwent sequential differentiation in two 60 % isopropanol baths (3 and 5 s), followed by rinsing in two distilled water baths (10 s each). They were then counterstained with hematoxylin (Servicebio, Wuhan, China) for 5 min and rinsed in three distilled water baths (5, 10, and 30 s). Differentiation was performed in a differentiation solution for 8 s, followed by rinsing in two distilled water baths (10 s each). The sections were blued in a bluing solution (Servicebio, Wuhan, China) for 1 s, rinsed in two tap water baths (5 and 10 s), and assessed for staining quality. Sections were mounted using a mounting medium and photographed microscopically.

2.2.4. Sirius red staining

Frozen sections were brought to room temperature, fixed in tissue fixative for 15 min, and rinsed with tap water. Sections were lightly dried, stained with Sirius Red A solution (Servicebio, Wuhan, China) for 2 min, rinsed twice with tap water, and then stained with Sirius Red B solution (Servicebio, Wuhan, China) for 30 min. Sections were dehydrated using absolute ethanol for four intervals of 5 s each, cleared with xylene for 1 min, and mounted with neutral balsam (Sinopharm Chemical Reagent Co., Ltd., Shanghai, China). Images were obtained with an optical microscope.

2.3. Biochemical analysis

Serum concentrations of TG, TC, LDL-C, HDL-C, SOD, and MDA were assessed using commercial kits from Nanjing Jiancheng Bioengineering Institute, Nanjing, China. Cell culture supernatants were assessed for SOD, MDA (Abbkine, Wuhan, China), and nitric oxide (NO) (Beyotime, Shanghai, China) using commercial kits.

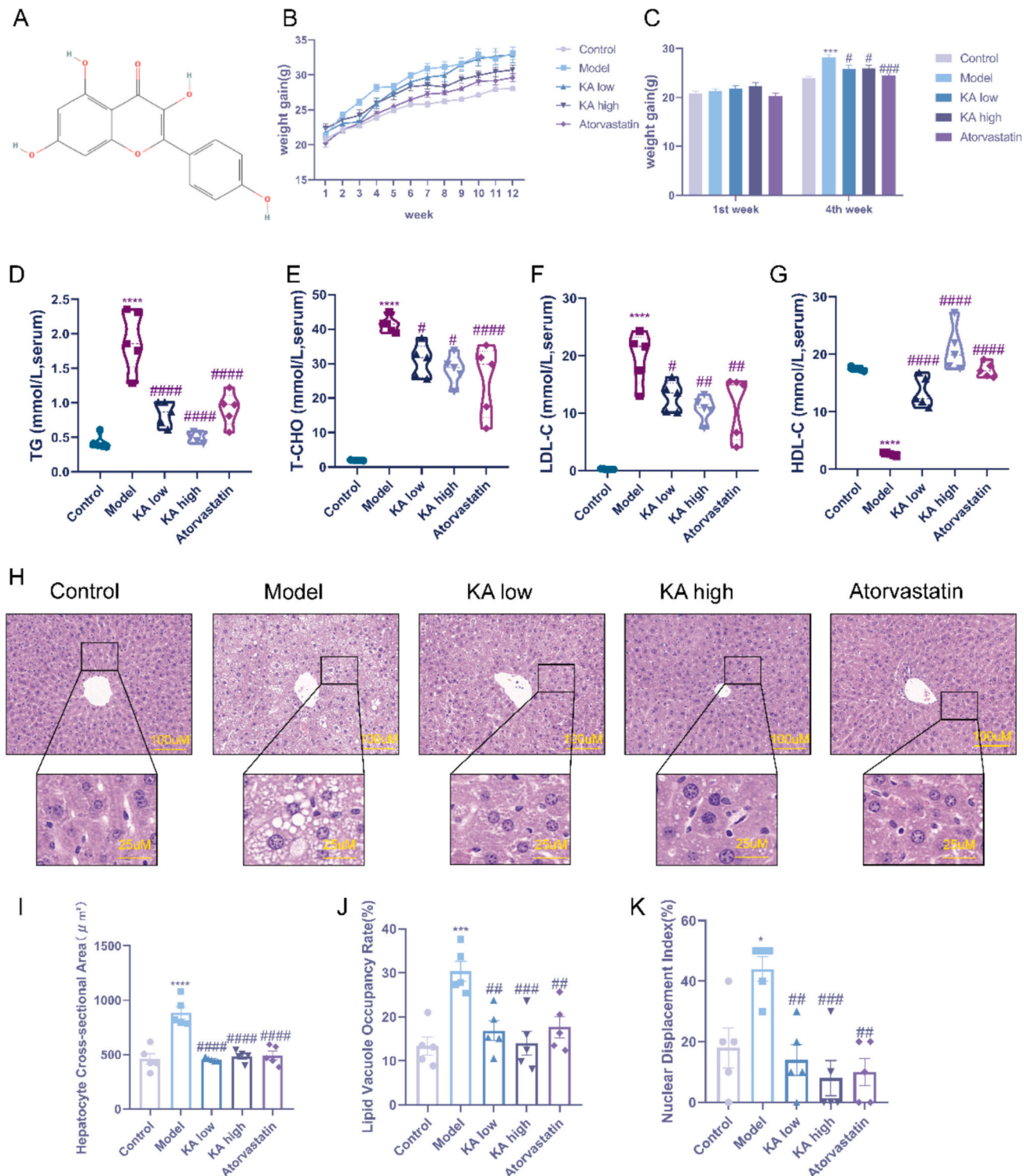


Fig. 1. KA modulates lipid metabolism and ameliorates hepatic steatosis in mice. (A) The chemical structure of KA can be found at PubChem(<https://pubchem.ncbi.nlm.nih.gov/>). (B) Body weight of mice in each group after 12 weeks of feeding on a HFD or standard diet. (C) Time course of body weight. Initial body weights were comparable among groups; significant differences emerged from week 4 onward. (unpaired *t*-test) (D) Serum triglyceride (TG) levels of each group after 12 weeks of feeding. (one-way ANOVA) (E) Serum total cholesterol (TC) levels of each group after 12 weeks of feeding. (one-way ANOVA) (F) Serum low-density lipoprotein cholesterol (LDL-C) levels of each group after 12 weeks of feeding. (one-way ANOVA) (G) Serum high-density lipoprotein cholesterol (HDL-C) levels of each group after 12 weeks of feeding. (one-way ANOVA) (H) Representative photomicrographs of liver tissue stained with hematoxylin and eosin (H&E) are provided. Low-magnification images (×40) illustrate overall histological appearance (scale bar = 100 μm), while high-magnification images (×250) reveal detailed cellular structures (scale bar = 25 μm). (I–K) Quantitative analysis of hepatocyte area (I), lipid vacuolization (J), and nuclear displacement (K) in H&E-stained liver sections. (one-way ANOVA) Significance levels: **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001 vs control group; #*p* < 0.05, ##*p* < 0.01, ###*p* < 0.001, *****p* < 0.0001 vs model group.

2.4. Immunofluorescence double labeling staining

Aortic root frozen sections underwent antigen retrieval with ice-cold solution (Solarbio, Beijing, China) for 10 min at room temperature, followed by permeabilization using 0.1 % Triton X-100 (BioFroxx, Munich, Germany) for 15 min, and were washed thrice with TBST (BioSharp, Wuhan, China). Sections were incubated with 10 % donkey serum (AntGene, Wuhan, China) at 37 °C for 30 min to block non-specific binding. Sections were incubated overnight at 4 °C with primary antibodies diluted in antibody diluent after removing the blocking serum. The next day, sections were brought to room temperature for 15 min and washed three times with TBST. Secondary antibodies, diluted at 1:400, were incubated at 37 °C for 45 min, followed by three washes with TBST. Nuclei were counterstained with DAPI working solution (1:500) (Solarbio, Beijing, China) for 5 min in the dark, washed once with TBST, and mounted with fluorescent mounting medium (Southern Biotech, Massachusetts, USA). Images were acquired with a fluorescence microscope from Olympus Corporation, Tokyo, Japan. The antibodies utilized were CD31 goat polyclonal (IgG, 1:100, R&D Systems, USA), Nrf2 rabbit polyclonal (IgG, 1:100, Bioss, China), Alexa Fluor®488 donkey anti-goat IgG (H + L) and Alexa Fluor®594 donkey anti-rabbit IgG (H + L) (both from Thermo Fisher, USA), Anti-P53, and Anti- γ -H2A.X (both 1:100, Abcam, UK).

2.5. Quantitative assessment of histological analyses

All stained sections (Oil Red O, Sirius Red, H&E, and Masson's Trichrome) were imaged using a high-resolution light microscope (model XYZ) with a digital camera, and immunofluorescence images were acquired on a confocal microscope. Imaging parameters (exposure, magnification, illumination, gain) were kept constant across samples.

Quantitative analysis was performed using ImageJ (NIH, Bethesda, MD, USA) by two independent investigators blinded to the treatment groups. For Oil Red O and Sirius Red/Masson's Trichrome staining, lipid- or collagen-positive areas were calculated as a percentage of the total lesion or vessel area after uniform thresholding. For H&E staining, intima-media thickness and foam cell area were measured. Masson's Trichrome analysis also included fibrous cap area, necrotic core area, cap/core ratio, rupture rate, and cap thickness. Liver histology was assessed for hepatocyte volume, lipid vacuoles, and nuclear displacement. For immunofluorescence, mean fluorescence intensity within defined ROIs was quantified after background subtraction; colocalization was assessed via Pearson's coefficient.

At least 3 sections per mouse and 5 mice per group were analyzed for each staining. Representative images were selected based on tissue quality and reproducibility. Data are expressed as mean $\pm$ SEM, with statistical analyses performed as described in the Statistical Analysis section.

2.6. RNA sequencing

RNA was extracted from aortic tissues of Control, Model, and KA high-dose groups ($n = 5$ each) using the MJZol Total RNA Extraction Kit (Shanghai Majorbio Bio-pharm Technology Co., Ltd., China). The NovaSeq X Plus platform was used for sequencing. Libraries were created using the SMART-Seq V4 Ultra Low Input RNA Kit for Sequencing. RSEM (<http://deweylab.github.io/RSEM/>) was used to quantify gene and transcript expression levels. Differentially expressed genes (DEGs) were identified using DESeq2 (<http://bioconductor.org/packages/stats/bioc/DESeq2/>) with criteria of $p \leq 0.05$ and $|\log_2 \text{fold change}| > 1.5$. GO enrichment analysis of DEGs was conducted with Goatools (<https://github.com/tanghaibao/GOatools>), while KEGG pathway enrichment analysis utilized KOBAS (<http://kobas.cbi.pku.edu.cn/home.do>). Subsequent visualization was performed on the Majorbio Cloud Platform [22].

2.7. Molecular docking

AutoDockTools (Version 1.5.7) was used to preprocess small molecules and proteins, select docking sites, and generate configuration files for AutoDock Vina. AutoDock Vina was employed for iterative local searches to identify optimal docking conformations. Analysis and visualization of the results were conducted using PyMOL (v3.0.3) and Maestro (v13.9.135).

2.8. Western blot

Aortic tissues or cultured cells were lysed with RIPA lysis buffer (XinsaiMei, Suzhou, China) containing a protease and phosphatase inhibitor cocktail. Protein quantification was performed using the BCA Protein Assay Kit (XinsaiMei, Suzhou, China). Protein samples were prepared using a protein loading buffer from XinsaiMei, Suzhou, China. Protein samples underwent SDS-PAGE electrophoresis and were transferred to a 0.22 μm pore-size nitrocellulose membrane (PVDF membrane, Millipore, Massachusetts, USA). The PVDF membrane was blocked with rapid blocking buffer at room temperature for 30 min, followed by incubation with primary antibodies at 4 °C overnight. Following five washes, the membrane was incubated with secondary antibodies at room temperature for 1 h, subjected to another five washes, and protein bands were visualized using an ECL chemiluminescence detection kit (Biosharp, Hefei, China). Results were processed and analyzed using ImageJ software. The study utilized the following primary antibodies: Anti-Heme Oxygenase 1 (1:1000, Abcam, UK), anti-eNOS (1:1000, Proteintech, China), anti-NQO1 (1:1000, Proteintech, China), anti-Nrf2 (1:1000, CST, USA), Anti-P53 (1:5000, Proteintech, China), Anti-p21 (1:1000, Abcam, UK), Anti-CDKN2A/p16INK4a (1:1000, Abcam, UK), anti-Beta Actin (1:20000, Proteintech, China), and anti-GAPDH (1:5000, Abcam, UK). Additionally, horseradish peroxidase (HRP)-labeled goat anti-rabbit IgG (H + L) and HRP-labeled goat anti-mouse IgG (H + L) were used, both at 1:1000 dilution from Beyotime, China. $n = 6$ biological replicates.

2.9. Cell culture and experimental design

Human umbilical vein endothelial cells (EA.hy926, Cat.No.SCSP-5285) were obtained from the Cell Bank of the Chinese Academy of Sciences and verified to be uncontaminated through STR profiling and mycoplasma testing. Cells were cultured in high-glucose DMEM (Nanjing SenBeiJia Biological Technology, China) with 10 % fetal bovine serum (Yeasen Biotechnology, China) and 1 % penicillin-streptomycin (Nanjing SenBeiJia Biological Technology, China) at 37 °C in a 5 % CO₂ humidified incubator (Thermo Scientific, Heracell 150i). Cells were passaged with 0.25 % trypsin-EDTA (Nanjing SenBeiJia Biological Technology, Nanjing, China) and seeded at a 1:3 ratio. All experiments were conducted using cells in the logarithmic growth phase (Passage 8–15). Experimental Groups: Control group: cultured in complete medium for 48 h. The model group was exposed to 300 μM H₂O₂ (Sino-pharm Chemical Reagent Co., Ltd., Shanghai, China) prepared fresh in PBS for 48 h. KA intervention group: co-treated with 300 μM H₂O₂ + 20/40 μM KA (MedChemExpress, New Jersey, USA) for 48 h. Mechanism validation groups: p53 inhibition group: co-treated with 300 μM H₂O₂ + 40 μM KA + 10 μM Pifithrin- μ (MedChemExpress, New Jersey, USA) for 48 h. P53 activation group: co-treated with 300 μM H₂O₂ + 40 μM KA + 80 μM Kevetrin hydrochloride (MedChemExpress, New Jersey, USA) for 48 h. Concentration screening was based on pre-experimental CCK-8 results (XinsaiMei, Suzhou, China), which were used to select non-cytotoxic concentrations of KA, H₂O₂, Pifithrin- μ , and Kevetrin hydrochloride.

2.10. Intracellular reactive oxygen species (ROS) assay

Cells were seeded in 6-well plates and pretreated with H₂O₂ and KA

(20 μ M, 40 μ M) for 48 h to evaluate intracellular ROS production. Cells were incubated with 10 μ M H2DCFDA (Biosharp, Hefei, China) at 37 °C in darkness for 30 min. Following two washes with serum-free medium, fluorescence intensity was examined using a fluorescence microscope.

2.11. CCK-8 assay

Cell suspensions were plated in 96-well plates and treated as necessary. Afterward, 10 μ L of CCK-8 reagent (XinsaiMei, Suzhou, China) was introduced to each well, and the plates were placed in a cell culture incubator for 2 h. Cell viability was determined by measuring absorbance at 450 nm with a microplate reader.

2.12. β -Galactosidase (SA- β -gal) staining

Cells were cultured on coverslips within 6-well plates and exposed to the necessary treatments. The medium was removed, followed by two PBS washes of the cells. Subsequently, 1 mL of fixative was introduced into each well. Following fixative removal, cells underwent three PBS washes before adding 1 mL of staining solution (Abbkine, Wuhan, China) to each well. The plates were incubated overnight at 37 °C without CO₂. The rate of SA- β -gal-positive cells was determined by observing them under a microscope.

2.13. EdU assay

Cells were cultured on coverslips within 6-well plates and exposed to the necessary treatments. Cells were incubated for an additional 2 h after adding pre-warmed (37 °C) 2 $\times$ EDU working solution (Beyotime, Shanghai, China) to the plates. After removing the medium, 1 mL of fixative was added to each well for 15 min. Cells were washed three times with 1 mL of washing buffer for 5 min each after the fixative was removed. Permeabilization buffer was applied to each well for 15 min, followed by three 5-min washes with washing buffer. Subsequently, 0.5 mL of Click reaction solution was introduced to each well, followed by a 30-min incubation at room temperature in the dark. The reaction solution was discarded, and 1 mL of 1 $\times$ Hoechst 33342 was introduced to each well for a 10-min incubation at room temperature in the dark. Cells were rinsed thrice with washing buffer, each for 5 min, and then examined using a fluorescence microscope. The Edu-positive cell rate was calculated.

2.14. Statistical analysis

Each experiment was conducted a minimum of three times. GraphPad Prism 8 (GraphPad Software Inc., CA, USA) was used for analyzing and plotting experimental results. Statistical analysis was performed using one-way ANOVA followed by Tukey's test or an unpaired *t*-test. A *p*-value below 0.05 was deemed statistically significant.

3. Results

3.1. KA improves systemic metabolism and plaque stability in AS models

3.1.1. KA regulates lipid metabolism and alleviates hepatic steatosis

3.1.1.1. Dynamic analysis of body weight and serum lipids. HFD-fed ApoE^{-/-} mice were used to induce AS, and the effects of two doses of KA on lipid metabolism were compared to both untreated and atorvastatin-treated controls. Five groups (*n* = 12 each) were studied: Control: C57BL/6 J mice on a normal diet; Model: ApoE^{-/-} mice on HFD; KA low: ApoE^{-/-} + HFD + 10 mg/kg/day KA; KA high: ApoE^{-/-} + HFD + 20 mg/kg/day KA; Atorvastatin: ApoE^{-/-} + HFD + 10 mg/kg/day atorvastatin. Dynamic monitoring of body weight (Fig. 1B-C) revealed that HFD-induced obesity emerged by week 3, with the model group

significantly heavier than controls (*p* < 0.01). After 4 weeks of KA treatment, both KA low- and high-dose groups exhibited significantly reduced weight gain relative to the model group (*p* < 0.05).

Serum lipid profiling (Fig. 2D-G) revealed significant dyslipidemia in HFD-fed ApoE^{-/-} mice compared to controls, with TG increasing 3.5-fold, TC 20-fold, LDL-C 141-fold (all *p* < 0.0001), and HDL-C decreasing by 85 % (*p* < 0.0001). KA administration produced dose-dependent lipoprotein improvements: KA low: TG ↓56 % (*p* < 0.0001), TC ↓26 % (*p* < 0.05), LDL-C ↓35 % (*p* < 0.05), HDL-C ↑4-fold (*p* < 0.0001); KA high: TG ↓74 % (*p* < 0.0001), TC ↓32 % (*p* < 0.05), LDL-C ↓45 % (*p* < 0.01), HDL-C ↑7-fold (*p* < 0.0001). Notably, the KA high-dose group outperformed the atorvastatin positive control in all parameters.

3.1.2. Pathological reversal of hepatic steatosis

Histological examination of liver sections (Fig. 2H-K) demonstrated pronounced steatosis in the model group: hepatocyte volume increased by 1.93-fold (*p* < 0.0001), lipid vacuoles occupied 30 % of the field (2.27-fold increase; *p* < 0.001), and nuclear displacement rose 2.44-fold (*p* < 0.05) versus controls. KA treatment improved morphology in a dose-dependent manner: at low doses, hepatocyte volume decreased by 49 % (*p* < 0.0001 vs. model), vacuole area by 45 % (*p* < 0.01), nuclear displacement by 68 % (*p* < 0.01); at high doses, hepatocyte volume, decreased by 46 % (*p* < 0.0001), vacuole area by 54 % (*p* < 0.001), nuclear displacement by 82 % (*p* < 0.001). Efficacy in the high-dose KA group exceeded that of atorvastatin.

3.1.3. KA inhibits aortic plaque formation and enhances plaque stability

3.1.3.1. Plaque area and lipid deposition. Oil Red O staining of the aortic sinus (Fig. 2A-B) revealed intact, lipid-free intima in control mice. In HFD-fed ApoE^{-/-} mice, lipid deposition increased by 41.0 % (*p* < 0.0001), with confluent plaques and luminal narrowing. KA treatment reduced plaque lipid cores in a dose-dependent fashion: KA low: plaque area ↓19.8 % (*p* < 0.05 vs. model); KA high: plaque area ↓34.6 % (*p* < 0.01); Atorvastatin: plaque area ↓36.1 % (*p* < 0.01). High-dose KA matched first-line clinical therapy (no significant difference vs. atorvastatin). H&E staining (Fig. 2C-E) further assessed plaque stability. The model group exhibited a 337.8-fold increase in intimal thickness (*p* < 0.0001) and foam cell infiltration constituting 29.2 % (*p* < 0.05) of plaque area. KA treatment tended to reduce foam cell content (20.9 % for KA low *p* = 0.870 vs. model, 22.8 % for KA high *p* = 0.947 vs. model), though differences versus the model did not reach statistical significance.

3.1.3.2. Collagen remodeling and fibrous cap enhancement. Sirius Red staining (Fig. 2F) showed markedly diminished collagen in the model group. KA high-dose restored collagen content close to control levels, suggesting reinforcement of the fibrous cap. Masson's trichrome staining (Fig. 2G-L) quantified microstructural changes: compared to controls, the model group displayed a fibrous cap area of 4.7 % (*p* < 0.001), necrotic core area of 16.1 % (*p* < 0.001), cap-to-core ratio of 33.0 % (*p* < 0.001), cap thickness of 20.64 μ m (*p* < 0.001), and elastic fiber breakage in 55.3 % of fields (*p* < 0.001). KA intervention significantly improved these metrics, with the high-dose group achieving: Fibrous cap area: 13.2 % of lumen (*p* < 0.01 vs. model); Necrotic core area: 5.9 % (*p* < 0.05); Cap-to-core ratio: 272.0 % (*p* < 0.001); Cap thickness: 36.28 μ m (*p* > 0.05); Elastic fiber integrity: breakage rate 20.8 % (*p* < 0.05).

Microscopic images were captured at 13 $\times$ magnification to display the overall histological changes of the aortic sinuses (scale bar = 400 μ m) and at 30 $\times$ magnification to reveal finer details (scale bar = 200 μ m). Significance levels: **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001 vs control group; #*p* < 0.05, ##*p* < 0.01, ###*p* < 0.001, ####*p* < 0.0001 vs model group. *n* = 5 biological replicates.

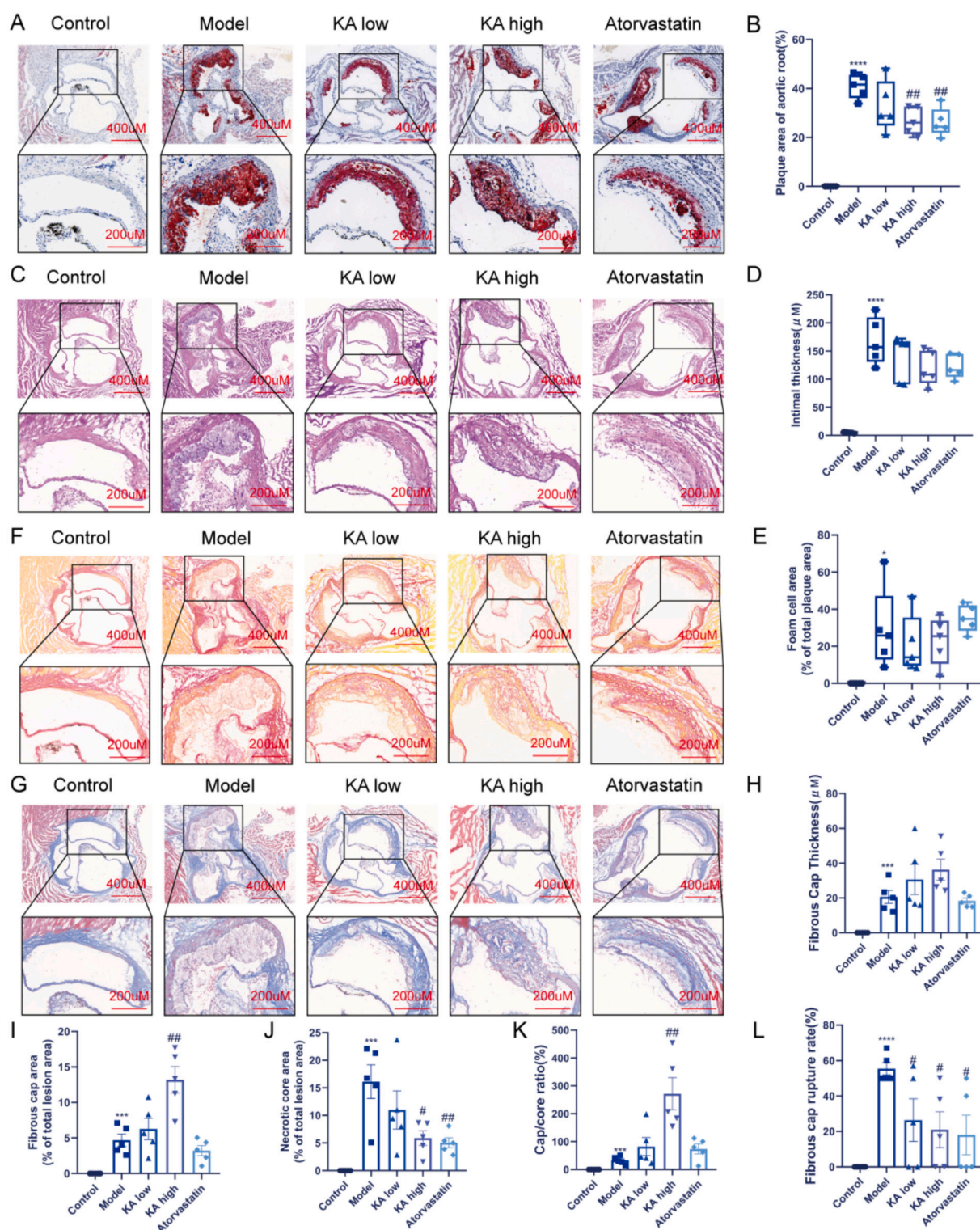


Fig. 2. Kaempferol reduces aortic plaque formation and improves plaque stability. (A) Illustrative images of Oil Red O-stained mouse aortas. (B) Quantitative assessment of plaque area within aortic sinuses. (unpaired *t*-test) (C) Representative images of mouse aortas stained with hematoxylin and eosin (H&E). (D) Quantitative assessment of intimal thickness in aortic sinuses. (unpaired *t*-test) (E) Percentage of foam cells within aortic sinus plaques. (unpaired *t*-test) (F) Representative images of Sirius Red-stained mouse aortas. (G) Representative Masson's Trichrome staining images of mouse aortas. (H) Quantitative assessment of fibrous cap thickness in aortic sinus plaques. (unpaired *t*-test) (I) Percentage of fibrous cap area in aortic sinus plaques. (unpaired *t*-test) (J) Percentage of necrotic core area in aortic sinus plaques. (unpaired *t*-test) (K) Cap-to-core ratio in aortic sinus plaques. (unpaired *t*-test) (L) Fibrous cap rupture rate (%) in aortic sinuses. (unpaired *t*-test).

3.2. KA Inhibits AS progression through dual mechanism of anti-oxidative stress and anti-cellular senescence

3.2.1. Molecular docking reveals dual binding modes of KA

3.2.1.1. Nrf2/KEAP1 interaction. Molecular docking predicted that KA binds with high affinity to the Nrf2 protein (binding energy -7.7 kcal/mol), forming multiple stabilizing contacts (Fig. 3A-B). In particular, KA establishes hydrogen bonds with conserved residues Arg43, Glu47, Lys50, Glu526, and Asp534, while its aromatic core occupies a hydrophobic pocket lined by Tyr46 and Val523. All interacting residues localize within Nrf2's Neh2 (KEAP1-binding) and Neh1 (DNA-binding) domains, suggesting that KA may both inhibit KEAP1-mediated ubiquitination and enhance Nrf2's recognition of antioxidant response elements (AREs).

3.2.1.2. p53 DNA-Binding domain blockade. Docking of KA to the p53 protein (binding energy -7.3 kcal/mol) revealed a distinct interaction pattern (Fig. 3C-D). KA interacts with Tyr327 through hydrogen bonding, π - π stacking, and van der Waals forces, securing its position within the DNA-binding domain. Additional polar interactions with Asn288 and Lys292 stabilize p53's DNA-binding interface, while hydrophobic contacts with Phe328 and Ala353 induce a slight displacement of the α -helix region, potentially hindering p53's association with its cognate DNA motifs. This unique occupancy implies an allosteric inhibitory mechanism.

3.2.2. KA mitigates oxidative stress in AS mice via activation of the Nrf2/HO1/NQO1 pathway

3.2.2.1. Transcriptomic analysis reveals regulation of antioxidant pathways. RNA-Seq of aortic tissues from control, model, and high-dose KA groups ($n = 5$ each) identified 4123 differentially expressed genes (DEGs; 1923 upregulated, 2200 downregulated) in HFD-fed ApoE^{-/-} mice versus controls ($p \leq 0.05$, $|\log_2FC| > 1.5$) (Fig. 4A-B). KA treatment reversed a subset of these changes, yielding 986 DEGs (307 upregulated, 679 downregulated) relative to model mice (Fig. 4C-D). The intersection

analysis identified 325 overlapping DEGs (Fig. 4E), which were enriched in PPAR and PI3K-Akt signaling pathways (Kyoto Encyclopedia of Genes and Genomes) (Fig. 4J), and positively associated with Nrf2-dependent antioxidant pathways (Gene Set Enrichment Analysis: KEAP1-NFE2L2 signaling, ROS detoxification, HMOX1-mediated cytoprotection) (Fig. 4F-I). The KA high-dose group showed a significant upregulation of Nrf2, HO1, and GPX8 expression (Fig. 4K-M), suggesting the restoration of endogenous antioxidant defenses.

3.2.2.2. KA dose-dependently activates the Nrf2/HO-1/NQO1 axis.

Western blot analysis of aortic lysates confirmed dose-dependent activation of the Nrf2 pathway (Fig. 5A-B). Compared to controls, HFD-fed model mice exhibited decreased levels of Nrf2 (-23% , $p < 0.01$), HO-1 (-53% , $p < 0.0001$), and NQO1 (-40% , $p < 0.05$). KA treatment induced the following changes versus model: KA low: NQO1 $+43\%$ ($p < 0.05$), Nrf2 $+22\%$ ($p = 0.13$), HO-1 $+39\%$ ($p = 0.07$); KA high: Nrf2 $+92\%$, HO-1 $+61\%$, NQO1 $+59\%$ (all $p < 0.05$), surpassing atorvastatin (Nrf2 $+77\%$, $p < 0.01$). These data demonstrate concentration-dependent enhancement of antioxidant signaling.

3.2.2.3. KA promotes endothelial protection via Nrf2 nuclear translocation.

Immunofluorescence of aortic root sections revealed that KA restores endothelial integrity by facilitating Nrf2 nuclear accumulation (Fig. 5C-D). In control mice, CD31 staining delineated a continuous endothelium with robust Nrf2 signal within nuclei. HFD-fed model mice showed disrupted CD31 continuity, reduced Nrf2 expression, and endothelial denudation. KA low-dose partially reinstated CD31 continuity and Nrf2 localization, whereas KA high-dose fully restored endothelial marker distribution and intensified nuclear Nrf2 accumulation.

3.2.2.4. KA mitigates serum oxidative stress in animal models.

Circulating oxidative stress markers corroborated KA's systemic antioxidant effects (Fig. 5E-F). Model mice exhibited a 1059% increase in MDA ($p < 0.001$) and a 34% decline in SOD activity ($p < 0.05$) relative to controls. KA intervention produced dose-dependent improvements: KA low: MDA -55% ($p < 0.05$), SOD $+61\%$ ($p < 0.01$); KA high: MDA -59% ($p < 0.05$), SOD $+64\%$ ($p < 0.01$), outperforming atorvastatin (MDA -50% ,

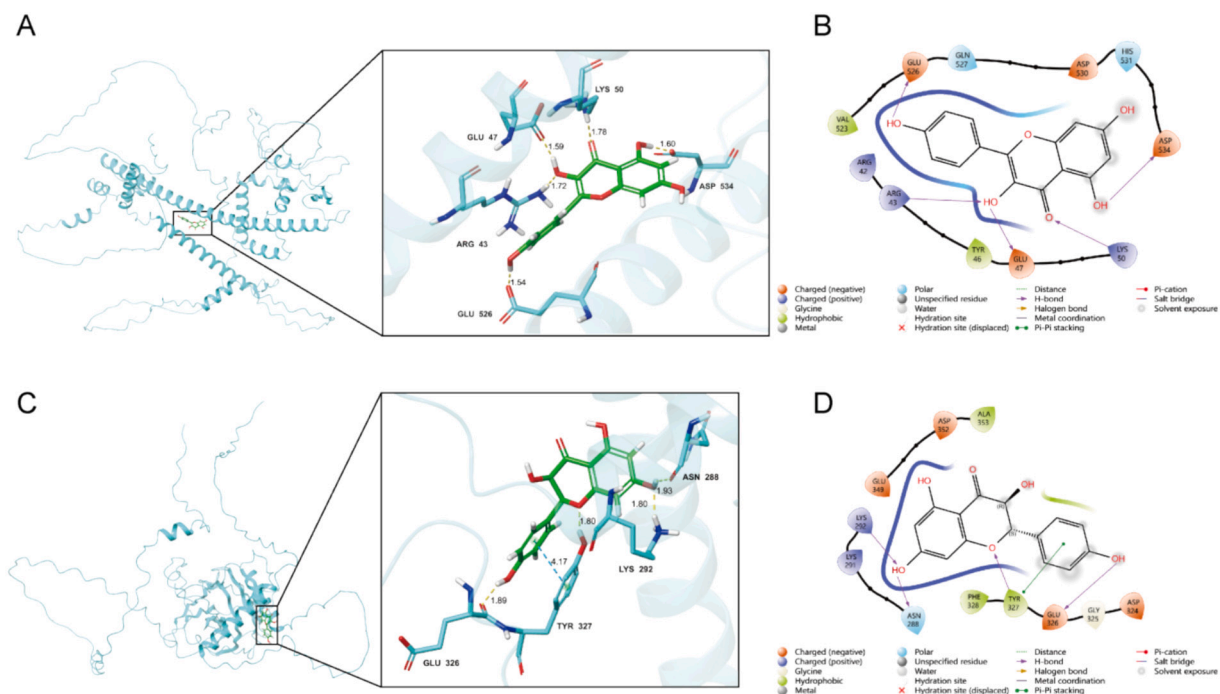


Fig. 3. Molecular Docking Reveals the Dual Binding Mode of KA. (A–B) Interaction diagram depicting the binding of KA with Nrf2. (C–D) Interaction diagram illustrating the binding of KA with p53.

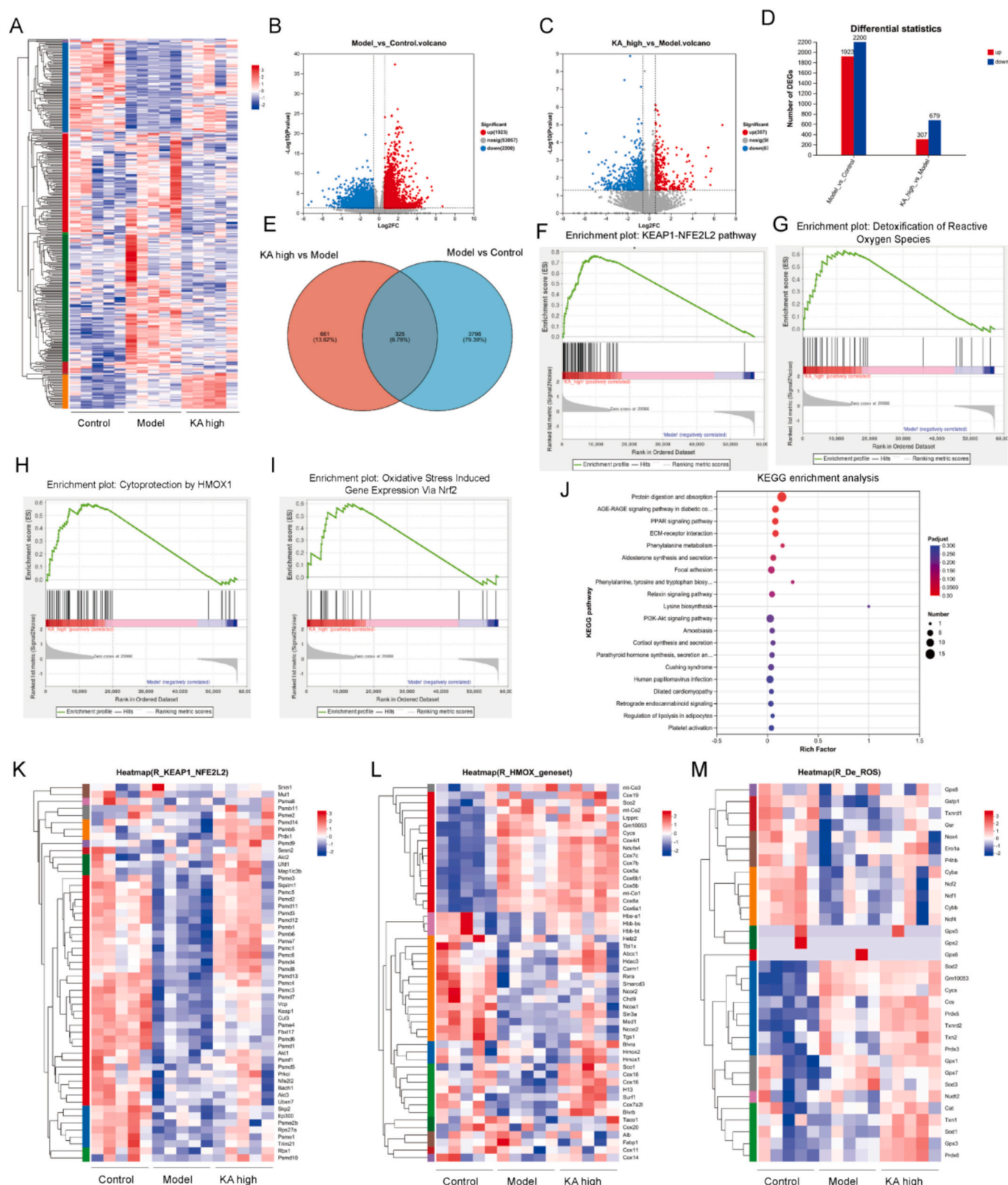


Fig. 4. Transcriptome Analysis Reveals the Regulatory Mechanism of KA's Antioxidant Pathway. (A) Heatmap illustrating changes in gene expression levels across the different experimental groups. (B–C) Volcano plots depicting differentially expressed genes (DEGs) from mouse aortic transcriptome data, where red denoting upregulation and blue indicating downregulation. (D) Bar plot analysis of DEGs derived from mouse aortic transcriptome data, highlighting upregulated genes in red and downregulated genes in blue. (E) Venn diagram illustrating the overlap of 325 shared DEGs between the KA vs. Model and Model vs. Control groups. (F–I) Gene set enrichment analysis (GSEA) identifies significant alterations in KEGG pathways linked to oxidative stress. (J) KEGG pathway analysis highlighting the enriched pathways of intersecting DEGs between KA vs. Model and Model vs. Control. (K–M) Heatmaps depicting gene expression patterns related to the KEAP1-Nrf2 pathway, the HO-1 pathway, and the ROS pathway. $n = 5$ biological replicates.

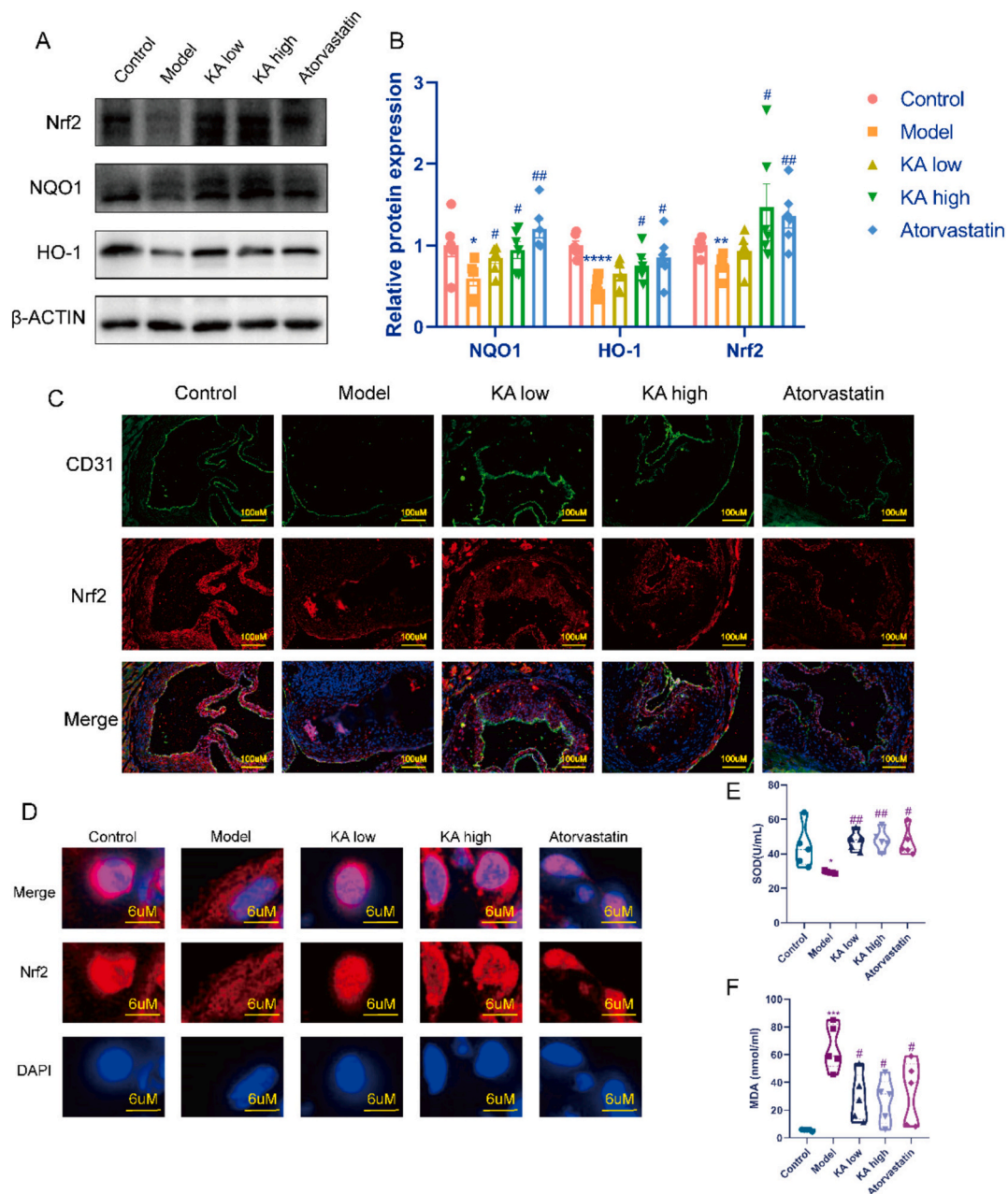


Fig. 5. Kaempferol Alleviates Oxidative Stress in Atherosclerotic Mice by Activating the Nrf2/HO-1/NQO1 Signaling Pathway. (A) Western blot analysis was conducted to assess the protein expression levels of Nrf2, NQO1, and HO-1 in the aortas of mice from each group. $n = 6$ biological replicates. (B) Relative quantification of protein expression levels for Nrf2, NQO1, and HO-1. (unpaired t -test) (C–D) Immunofluorescence double staining for the endothelial cell-specific marker CD31 (green) and the antioxidant stress factor Nrf2 (red) in the aortic sinus of mice from each group. Microscopic images were captured at $25\times$ magnification to display histological changes of the entire plaque (scale bar = $100\ \mu\text{m}$), with $584\times$ magnification used to reveal finer details (scale bar = $6\ \mu\text{m}$). $n = 5$ biological replicates. (E) Serum MDA levels in mice across different groups. $n = 5$ biological replicates. (one-way ANOVA) (F) Serum SOD activity in mice across different groups. $n = 5$ biological replicates. (one-way ANOVA) Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ vs control group; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$, #### $p < 0.0001$ vs model group.

$p < 0.05$; SOD +59 %, $p < 0.05$). KA high-dose further modestly enhanced SOD and reduced MDA relative to KA low, indicating a slight dose-response effect.

3.2.3. KA attenuates atherosclerotic plaque vulnerability by inhibiting endothelial cell senescence

3.2.3.1. KA suppresses DNA damage-induced endothelial senescence. The antisenescent properties of KA on vascular endothelium were evaluated by measuring $\gamma\text{H2A.X}$, a DNA damage marker, using

immunofluorescence in aortic root sections (Fig. 6A–B). Compared with the control group, HFD-fed ApoE^{−/−} mice displayed an 88.2 % increase in $\gamma\text{H2A.X}$ fluorescence intensity ($p < 0.05$), indicating accumulation of DNA damage-induced senescent cells within plaques. KA treatment markedly attenuated this signal: fluorescence decreased by 86.9 % in the KA low group and by 89.3 % in the KA high group (both $p < 0.0001$ vs. model), exceeding the reduction observed with atorvastatin (−64.2 %, $p < 0.05$ vs. KA high).

Co-localization of p53 (senescence marker) and CD31 (endothelial marker) revealed endothelial integrity and senescence status (Fig. 6C–E).

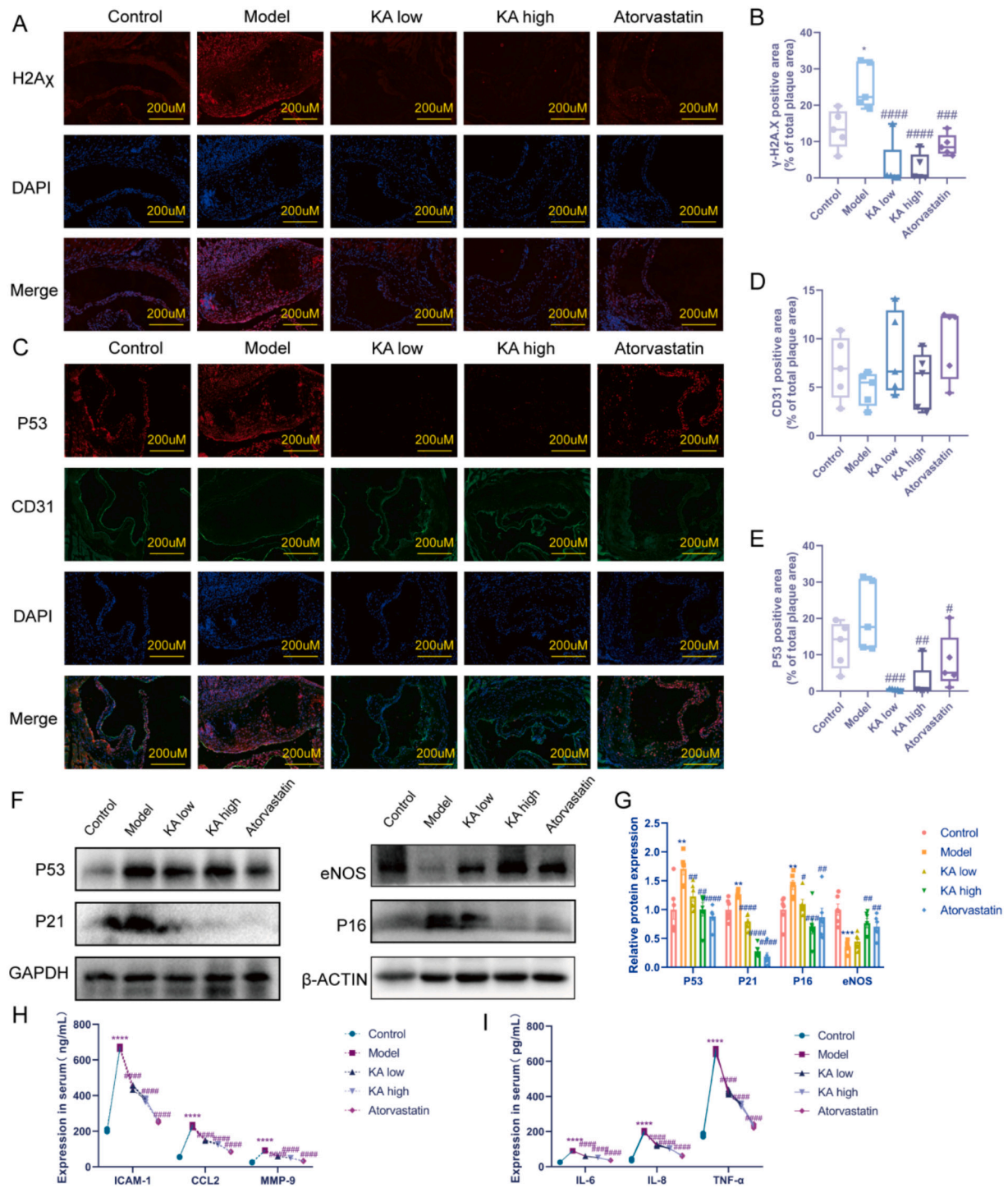


Fig. 6. Kaempferol Alleviates Atherosclerotic Plaque Vulnerability by Inhibiting Endothelial Cell Senescence. (A) Immunofluorescence staining of γ -H2A.X (red), a DNA damage marker, was performed on aortic sinus sections from each mouse group. Histological changes in plaque regions were observed in microscopic images captured at 30 $\times$ magnification (scale bar = 200 μ m). $n = 5$ biological replicates. (B) Relative quantification of γ -H2A.X fluorescence intensity. (one-way ANOVA) (C) Double immunofluorescence staining was performed on aortic sinus sections from each mouse group, targeting the cellular senescence marker p53 (red) and the endothelial cell-specific marker CD31 (green). Histological changes in plaque regions were observed in microscopic images captured at 30 $\times$ magnification (scale bar = 100 μ m). $n = 5$ biological replicates. (D) Relative quantification of CD31 fluorescence signals. $n = 5$ biological replicates. (one-way ANOVA) (E) Relative quantitative analysis of p53 fluorescence signals was conducted. (one-way ANOVA) (F) Western blot analysis was conducted to assess the expression levels of p53, p21, p16, and eNOS proteins in the aortas of mice across different groups. $n = 6$ biological replicates. (G) Relative quantitative analysis of p53, p21, p16, and eNOS protein expression levels. (unpaired t -test) (H–I) Expression levels of senescence-associated secretory phenotype (SASP) factors, including IL-6, IL-8, TNF- α , ICAM-1, CCL2, and MMP-9, in serum samples from each group. $n = 5$ biological replicates. (one-way ANOVA) $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, $^{****}p < 0.0001$ vs control group; $^{#}p < 0.05$, $^{##}p < 0.01$, $^{###}p < 0.001$, $^{####}p < 0.0001$ vs model group.

In controls, CD31-positive cells formed a continuous monolayer (7.0 % positive rate) and p53 expression was minimal (12.7 %). HFD-fed mice exhibited disrupted CD31 staining (4.9 %, $p = 0.85$ vs. control) and increased p53 positivity (20.7 %, $p = 0.33$ vs. control). KA intervention restored endothelial continuity (CD31: $p = 0.47$ vs. control) and dramatically suppressed p53 expression to 0.4 % in the KA low group ($p < 0.01$ vs. model), outperforming atorvastatin (8.0 %, $p = 0.38$ vs. KA low).

3.2.3.2. KA modulates the p53-p21-p16/eNOS pathway to inhibit senescence. Western blot analysis of aortic lysates demonstrated that HFD induced significant upregulation of senescence-associated proteins—p53 (+71.0 %), p21 (+27.0 %), p16^{INK4a} (+43.0 %; all $p < 0.01$)—and downregulation of endothelial nitric oxide synthase (eNOS: −64.0 %, $p < 0.001$) relative to controls (Fig. 6F–G). KA treatment reversed these changes in a dose-dependent manner: KA low: p53−28.0 % ($p < 0.01$), p21−38.0 % ($p < 0.0001$), p16−23.0 % ($p < 0.05$), eNOS+24.0 % ($p = 0.23$); KA high: p53−41.0 % ($p < 0.01$), p21−78.0 % ($p < 0.0001$), p16−50.0 % ($p < 0.001$), eNOS+112.0 % ($p < 0.01$). Atorvastatin produced similar reductions in senescence markers and eNOS restoration, with no significant differences compared to KA high ($p > 0.05$ for all comparisons), confirming that KA effectively modulates the p53-p21-

p16/eNOS axis to inhibit endothelial senescence.

3.2.3.3. KA antagonizes the senescence-associated secretory phenotype (SASP). To evaluate SASP modulation, we measured serum concentrations of pro-inflammatory and matrix-remodeling factors by ELISA (Fig. 6H–I). HFD-fed mice exhibited marked increases in IL-6 (+259 %), IL-8 (+392 %), TNF- α (+269 %), ICAM-1 (+230 %), CCL2 (+313 %), and MMP-9 (+270 %) compared to controls (all $p < 0.0001$). KA markedly suppressed these SASP components in a dose-dependent fashion: KA low: IL-6 −82 %, IL-8 −87 %, TNF- α −82 %, ICAM-1 −79 %, CCL2−84 %, MMP-9 −83 % (all $p < 0.0001$ vs. model); KA high: IL-6 −84 %, IL-8 −89 %, TNF- α −85 %, ICAM-1 −83 %, CCL2−86 %, MMP-9 −46 % (all $p < 0.0001$). Atorvastatin achieved comparable reductions for most factors, but more potently decreased MMP-9 (−90 % vs. −46 % for KA high, $p = 0.018$), indicating that KA selectively modulates SASP to improve the vascular microenvironment.

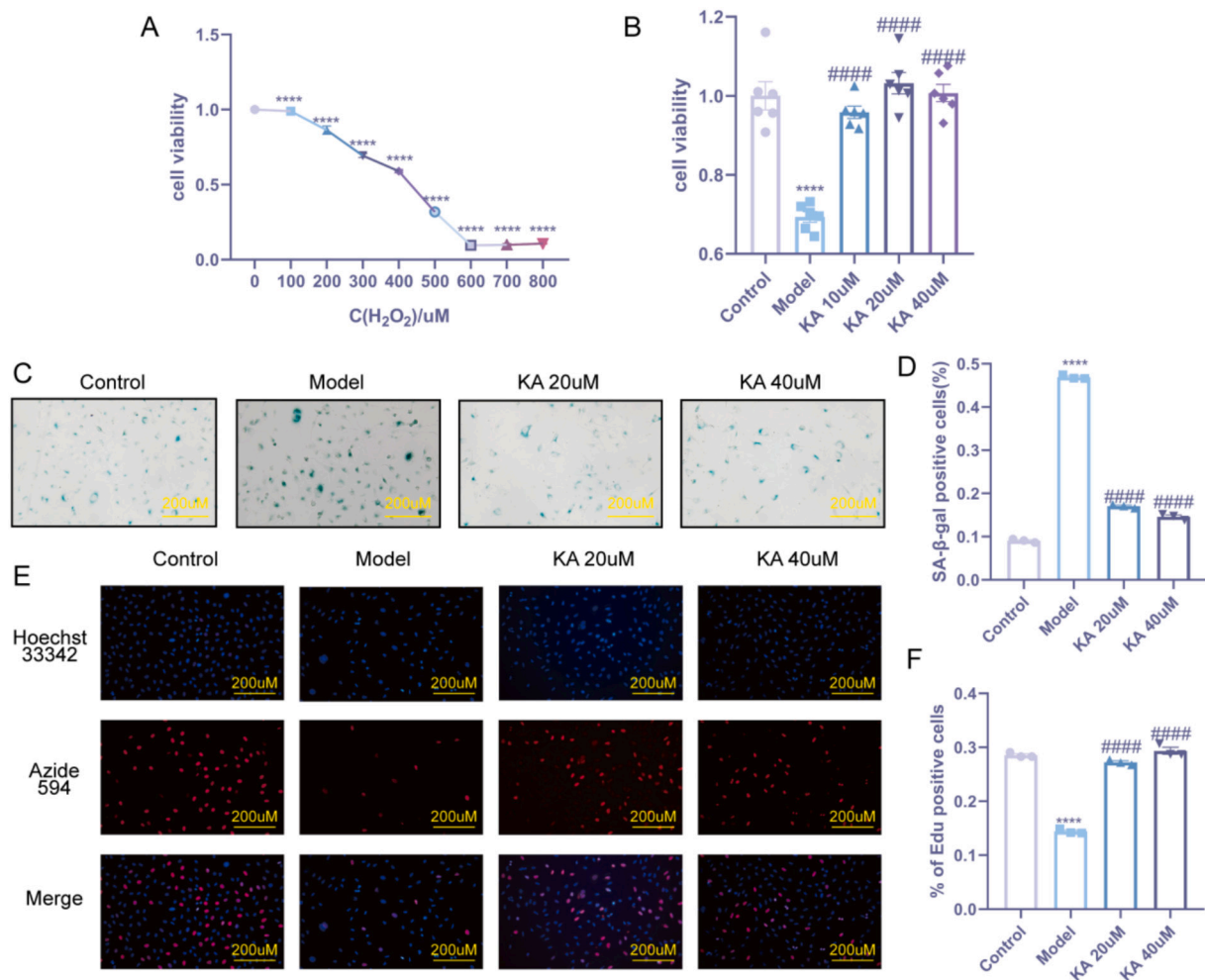


Fig. 7. Kaempferol Inhibits H_2O_2 -Induced Endothelial Cell Senescence Phenotypes and Restores Functionality. (A) The impact of varying H_2O_2 concentrations on the viability of endothelial cells. $n = 6$ biological replicates. (one-way ANOVA) (B) The rescue effect of Kaempferol (KA) on H_2O_2 -induced endothelial cell viability. $n = 6$ biological replicates. (one-way ANOVA) (C) Representative SA- β -gal stained images for each group, scale bar = 200 μm . $n = 3$ biological replicates. (D) Relative quantification of the SA- β -gal positive rate in each group. (one-way ANOVA) (E) EdU staining images for each group are shown, with a scale bar of 200 μm . Azide 594 fluorescence stand for the labeling of EdU positive cells. $n = 3$ biological replicates. (F) Relative quantification of the EdU positive rate was performed for each group. (one-way ANOVA) * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ vs control group; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$, #### $p < 0.0001$ vs model group.

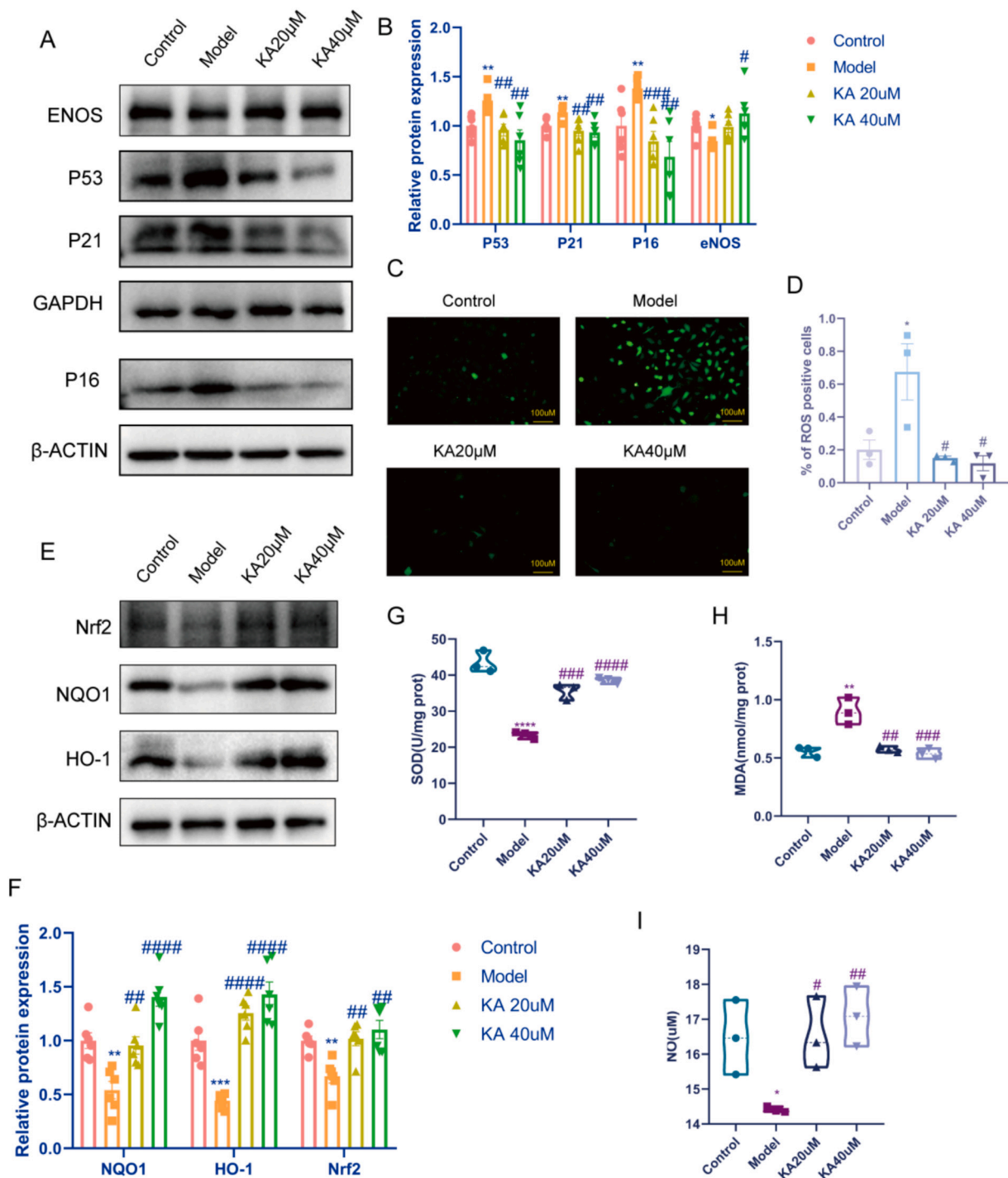


Fig. 8. Kaempferol Mitigates Oxidative Stress Damage in Endothelial Cells Induced by H_2O_2 . (A) Western blot analysis was conducted to assess the protein expression levels of p53, p21, p16, and eNOS in endothelial cells across different experimental groups. $n = 6$ biological replicates. (B) Relative quantitative analysis of the protein expression levels of p53, p21, p16, and eNOS. (unpaired t-test) (C) Representative images depicting reactive oxygen species (ROS) levels across different cell groups. Scale bar = 100 μ m. $n = 3$ biological replicates. (D) Relative quantification of ROS levels across different cell groups. (E) Western blot analysis was conducted to assess the protein expression levels of Nrf2, NQO1, and HO-1 in endothelial cells from each group. $n = 6$ biological replicates. (F) Relative quantitative analysis of protein expression levels of Nrf2, NQO1, and HO-1. (unpaired t-test) (G) Supernatant superoxide dismutase (SOD) activity in each group of cells. $n = 3$ biological replicates. (one-way ANOVA) (H) Malondialdehyde (MDA) activity in the supernatant of each group of cells. $n = 3$ biological replicates. (one-way ANOVA) (I) Nitric oxide (NO) activity in the supernatant of each group of cells. $n = 3$ biological replicates. (unpaired t-test) * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ vs control group; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$, #### $p < 0.0001$ vs model group.

3.3. Mechanistic validation of KA-Mediated inhibition of endothelial cell senescence via the p53 pathway

3.3.1. KA suppresses H₂O₂-INDUCED endothelial cell senescence phenotypes and restores function

Senescence was induced in EA.hy926 cells by treatment with 300 μ M

H₂O₂(Fig. 7A), as evidenced by a 4.18-fold increase in SA- β -gal-positive cells and a 50 % reduction in EdU incorporation (both $p < 0.0001$) relative to untreated controls, confirming oxidative stress-mediated cell cycle arrest. Through CCK8 assay screening, 20 and 40 μ M KA were selected as the intervention concentrations(Fig. 7B). Co-treatment with KA (20 and 40 μ M) dose-dependently reduced SA- β -gal positivity by 64

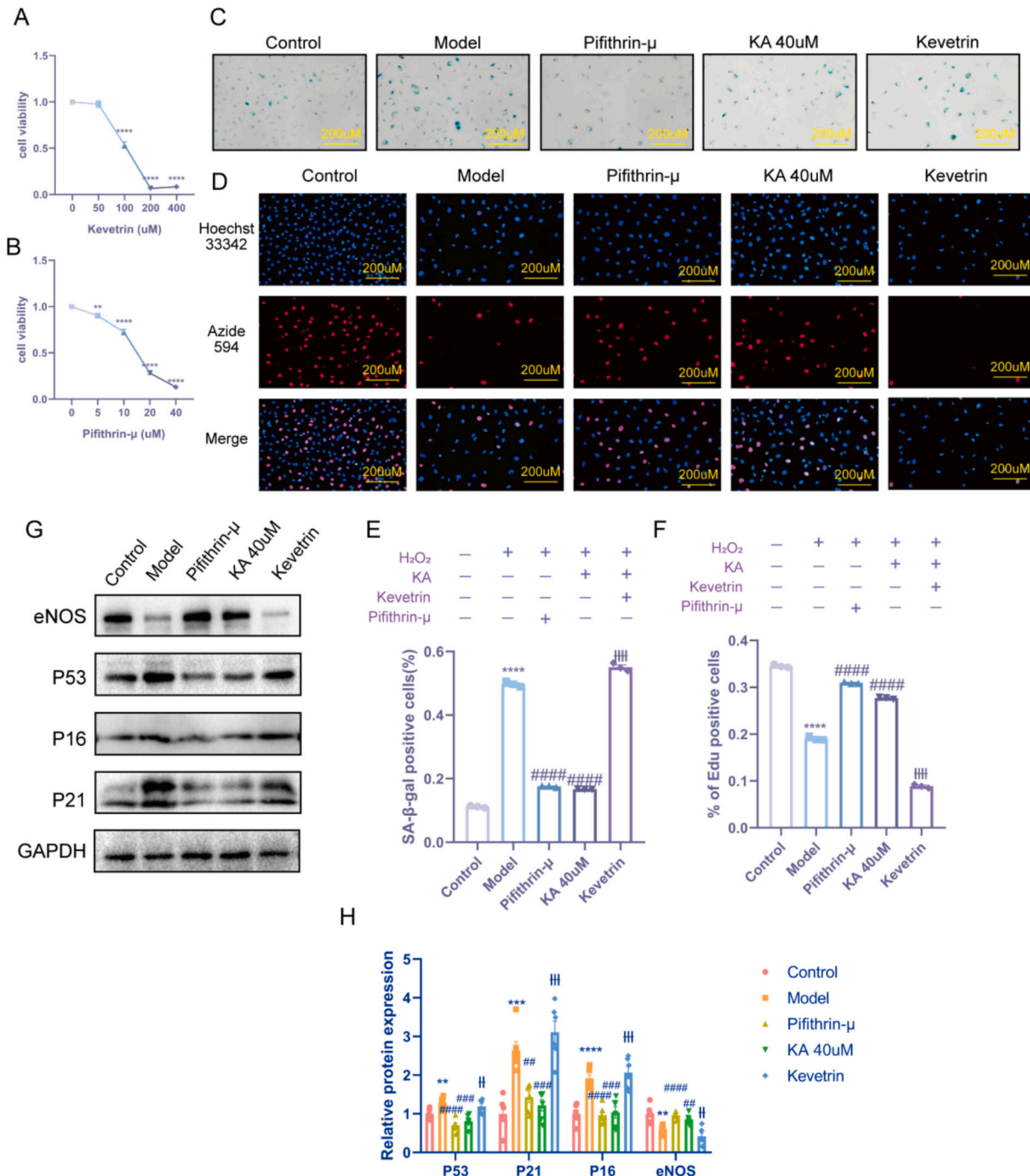


Fig. 9. Functional Validation and Mechanistic Analysis of the p53 Signaling Pathway. (A) Impact of different Kevetrin concentrations on the viability of endothelial cells. $n = 6$ biological replicates. (one-way ANOVA) (B) Impact of varying Pifithrin- μ concentrations on endothelial cell viability. $n = 6$ biological replicates. (one-way ANOVA) (C) Representative images of SA- β -gal staining for each group after combined treatment with the p53 inhibitor (Pifithrin- μ) and activator (Kevetrin). Scale bar = 200 μ m. $n = 3$ biological replicates. (D) Representative images of EdU staining in each group following combined treatment with the p53 inhibitor (Pifithrin- μ) and activator (Kevetrin). Scale bar = 200 μ m. $n = 3$ biological replicates. (E) Relative quantification of the SA- β -gal positive rate in each group. (one-way ANOVA) (F) Relative quantification of the EdU positive rate in each group. (one-way ANOVA) (G) Western blot analysis of p53, p21, p16, and eNOS protein expression levels in each group after combined treatment with the p53 inhibitor (Pifithrin- μ) and activator (Kevetrin). $n = 6$ biological replicates. (H) Relative quantitative analysis of p53, p21, p16, and eNOS protein expression levels. (unpaired t -test) * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ vs control group; # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$, #### $p < 0.0001$ vs model group; $^{\dagger}p < 0.05$, $^{\dagger\dagger}p < 0.01$, $^{\dagger\dagger\dagger}p < 0.001$, $^{\dagger\dagger\dagger\dagger}p < 0.0001$ vs KA 40 μ M group.

% and 69 % (both $p < 0.0001$) and restored EdU incorporation to 89 % and 104 % of control levels (both $p < 0.0001$), indicating effective reversal of senescence-associated functional impairments (Fig. 7C-F).

Western blot analysis corroborated these findings: H_2O_2 significantly upregulated p53 (+26 %), p21 (+14 %), and p16^{INK4a} (+38 %; all $p < 0.01$) while decreasing eNOS expression by 15 % ($p < 0.05$) (Fig. 8A-B). KA at 20 μM downregulated p53 by 24 % ($p < 0.01$), p21 by 16 % ($p < 0.01$), and p16^{INK4a} by 39 % ($p < 0.001$), and increased eNOS by 17 % ($p = 0.059$). At 40 μM , KA further suppressed p53 (−32 %), p21 (−18 %), and p16^{INK4a} (−50 %; all $p < 0.01$) and elevated eNOS by 33 % ($p < 0.05$), demonstrating that KA preserves endothelial function by modulating key cell-cycle regulators and nitric oxide synthesis.

3.3.2. KA mitigates H_2O_2 -induced oxidative stress in endothelial cells

Exposure to 300 μM H_2O_2 significantly increased oxidative damage in EA.hy926 cells, with a 3.35-fold rise in intracellular ROS ($p < 0.05$), a 61 % increase in MDA ($p < 0.01$) (Fig. 8C-D), a 46 % reduction in SOD activity ($p < 0.0001$), and a 13 % decrease in NO production ($p < 0.05$) compared to controls (Fig. 9G-I). KA at 20 μM attenuated ROS accumulation by 77.7 % ($p < 0.05$), reduced MDA by 36 % ($p < 0.01$), enhanced SOD activity by 52 % ($p < 0.001$), and increased NO by 15 % ($p < 0.05$). At a concentration of 40 μM , KA further suppressed ROS (−82.4 %, $p < 0.05$) and MDA (−40 %, $p < 0.001$) while augmenting SOD (+65 %, $p < 0.001$) and NO (+19 %, $p < 0.01$), confirming dose-dependent protection against oxidative stress.

Consistent with these functional assays, Western blotting demonstrated that H_2O_2 suppressed the Nrf2 antioxidant axis—Nrf2 (−33 %, $p < 0.01$), NQO1 (−46 %, $p < 0.001$), and HO-1 (−56 %, $p < 0.01$) (Fig. 8E-F). KA at 20 μM upregulated Nrf2 by 52 % ($p < 0.01$), NQO1 by 77 % ($p < 0.01$), and HO-1 by 184 % ($p < 0.0001$). At 40 μM , KA further elevated Nrf2 (+65 %, $p < 0.01$), NQO1 (+160 %, $p < 0.0001$), and HO-1 (+223 %, $p < 0.0001$), confirming that KA enhances endothelial antioxidant defenses via Nrf2 pathway activation.

3.3.3. Functional validation and mechanistic analysis of the P53 signaling pathway

To confirm p53 dependence, endothelial cells were co-treated with the p53 inhibitor Pifithrin- μ (10 μM) or activator Kevetrin (80 μM) alongside KA (Fig. 9A-B). Pifithrin- μ mirrored KA's anti-senescence effects, reducing SA- β -gal positivity by 65 % ($p < 0.0001$) and restoring EdU incorporation by 63 % ($p < 0.0001$). In contrast, Kevetrin abrogated KA's protective action, increasing SA- β -gal positivity by 230 % ($p < 0.0001$ vs. KA 40 μM) and decreasing EdU incorporation by 68 % ($p < 0.0001$), demonstrating that KA's reversal of endothelial senescence requires p53 modulation (Fig. 9C-F).

Western blot analysis of p53 pathway proteins further substantiated these results: Pifithrin- μ suppressed p53 (−46 %, $p < 0.0001$), p21 (−46 %, $p < 0.01$), and p16^{INK4a} (−49 %, $p < 0.0001$) while upregulating eNOS (+58 %, $p < 0.0001$), paralleling the effects of KA 40 μM . Conversely, Kevetrin reversed KA's regulation—elevating p53 (+48 %), p21 (+155 %), and p16^{INK4a} (+99 %; all $p < 0.01$)—thereby validating that KA exerts its endothelial anti-senescence effects through modulation of the p53 signaling axis (Fig. 9G-H).

4. Discussion

This study provides the first comprehensive demonstration that KA, a natural flavonoid, impedes AS progression by concurrently modulating the cell senescence and oxidative-stress regulatory axes in both in vivo and in vitro models. Our findings deepen the mechanistic understanding of flavonoid-mediated cardioprotection and introduce a “multi-target synergistic” paradigm that may overcome the limitations of monotherapy approaches.

4.1. Mechanistic Innovation: Integrative Regulation of Senescence and Oxidative Stress

AS pathogenesis is driven by a deleterious feedback loop in which ROS induce endothelial DNA damage and cell senescence, while SASP factors (e.g., IL-6, MMP-9) exacerbate inflammation and oxidative injury [23]. Our data reveal that KA disrupts this cycle via dual mechanisms. First, KA activates the Nrf2/HO-1/NQO1 antioxidant pathway, evidenced by marked reductions in aortic MDA and restoration of SOD activity [24–27]. Second, KA inhibits of the p53/p21/p16 senescence cascade, resulting in fewer SA- β -gal-positive cells and attenuation of SASP-mediated inflammation (Fig. 10). This bidirectional control establishes a molecular “Redox-Senescence Homeostasis,” underscoring the therapeutic potential of simultaneously targeting redox balance and senescence in multifactorial diseases such as AS [28,29]. Interestingly, our findings suggest that KA's vascular protective actions are mediated through both independent and interconnected regulation of the Nrf2 and p53 pathways. Molecular docking and expression analyses indicate that KA can directly activate Nrf2, thereby inducing HO-1/NQO1 transcription and restoring redox homeostasis. In parallel, KA suppresses the p53-p21-p16 senescence axis, alleviating endothelial cell cycle arrest and dysfunction. This dual targeting may be functionally synergistic: previous studies have shown that excessive p53 activation can suppress Nrf2 transcription and negatively regulates antioxidant activity [30]. The mutant p53 (including R175H) gains function, leading to interaction with Nrf2, promoting its nuclear translocation and the activation of antioxidant signaling proteins in the cell [31]. Thus, KA's inhibition of p53 not only prevents endothelial senescence but may also indirectly sustain Nrf2 activation by relieving this inhibitory cross-talk. Together, these actions establish a “redox-senescence” balance in the vascular microenvironment, which may be essential for stabilizing atherosclerotic plaques and improving endothelial integrity.

Interestingly, low-dose KA (10 mg/kg) elicited a more pronounced upregulation of CD31 than the higher dose (20 mg/kg), suggesting that endothelial protective effects may follow a non-linear dose–response pattern. This phenomenon may relate to KA's role in mitigating endothelial cell senescence. Excessive flavonoid concentrations have been reported to exert paradoxical pro-oxidant effects or induce mild metabolic stress, which could attenuate endothelial regeneration and junctional protein expression [32,33]. At the lower dose, KA may optimally activate the Nrf2 antioxidant axis and suppress p53-mediated senescence without triggering compensatory stress responses, thereby preserving endothelial integrity and maintaining CD31 expression. Such a hormetic profile is consistent with prior studies demonstrating that modest activation of cytoprotective pathways can yield superior vascular outcomes compared to higher-dose interventions. These findings underscore the therapeutic importance of precise dose optimization to maximize endothelial rejuvenation while avoiding potential dose-related attenuation.

4.2. Theoretical Breakthrough: Expanding the Concept of Plaque Stability.

Conventional AS therapies focus primarily on lipid lowering, yet many patients show plaque progression despite optimal LDL reduction [4]. Our histopathological analyses demonstrate that high-dose KA increases the fibrous cap/necrotic core ratio by 7.24-fold ($p < 0.01$) and enhances collagen deposition—changes strongly correlated with p53 inhibition (−41 %, $p < 0.01$) and reduced foam cell content (−21.7 %, n.s.) [15,16,34]. These observations redefine plaque stability as dependent not only on lipid-core reduction but also on the dynamic interplay between cellular senescence and extracellular matrix remodeling. By promoting collagen synthesis and reinforcing cap integrity, KA exemplifies a novel “matrix metabolic reprogramming” approach to plaque stabilization [35,36].

By contrast, the reduction in macrophage-derived foam cell area

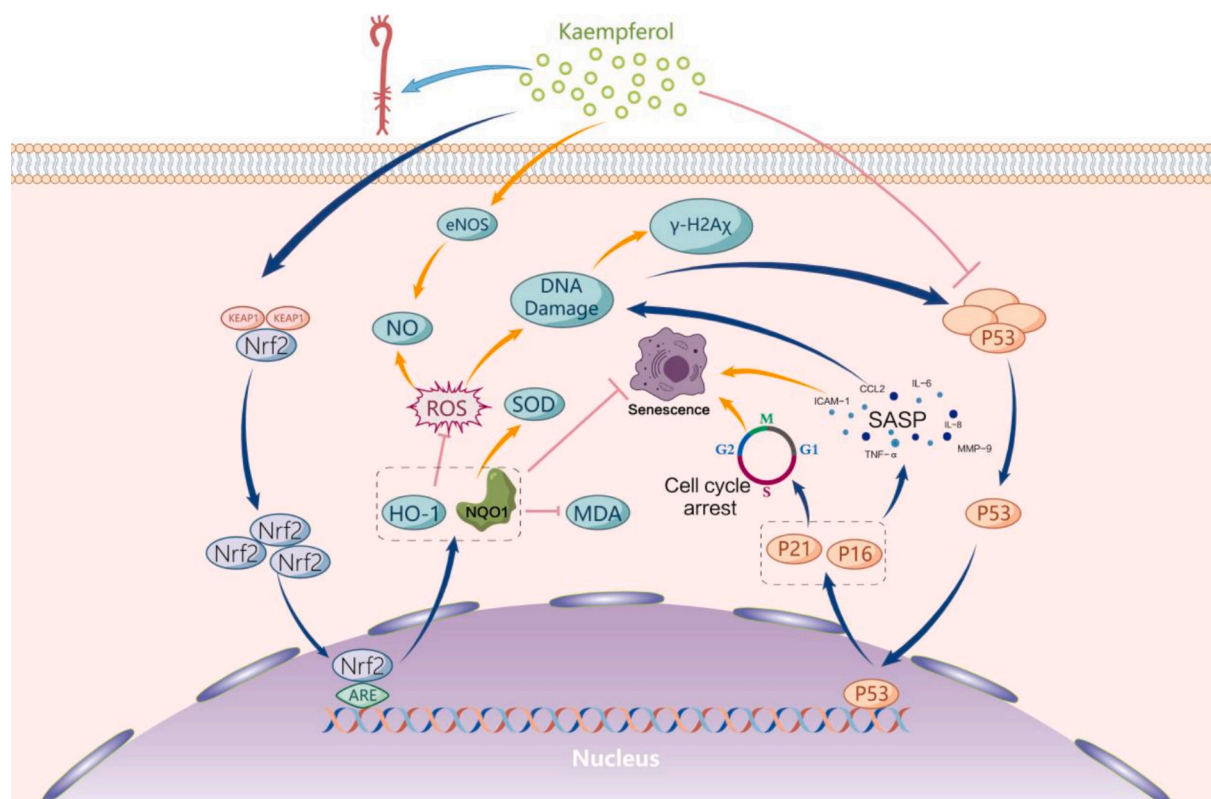


Fig. 10. Kaempferol's Dual Suppression of Vascular Oxidative Damage and Endothelial senescence.

(Fig. 2E-F) did not reach statistical significance. This discrepancy may simply reflect limited statistical power—our animal cohorts were relatively small, so a moderate biological effect on foam-cell burden could escape significance—rather than a failure of KA to act. It is also biologically plausible that KA's primary action is on vascular remodeling rather than lipid handling: by preserving endothelial health and reducing local inflammation, KA likely promotes smooth muscle cell-mediated matrix deposition (thickening the fibrous cap) without directly accelerating macrophage cholesterol efflux [37]. In other words, KA may “stabilize” plaques more so than it reduces lipid core size. Notably, even the observed downward trend in foam cells is directionally consistent with plaque stabilization, since abundant foam cells enlarge the necrotic core and weaken the plaque. Thus, although the foam cell reduction did not achieve $p < 0.05$, the overall pattern—stronger caps and less foam burden—points to a biologically meaningful benefit. Future studies with larger sample sizes or longer treatment will be needed to determine whether these trends in foam-cell content become significant, but the current results already support a potent plaque-stabilizing effect of KA in keeping with its anti-aging mechanism.

4.3. 3. Clinical implications: integrating natural flavonoids into precision cardiovascular therapy

While statins remain the cornerstone of AS management, their benefits are largely lipid-centric [5,38], with limited direct modulation of endothelial senescence or oxidative stress. Residual cardiovascular risk remains significant, and long-term use is associated with adverse metabolic effects, including increased incidence of new-onset diabetes [39]. Senolytic agents (e.g., dasatinib/quercetin) selectively clear senescent cells but do not enhance antioxidant defenses or stabilize the extracellular matrix, and their safety in chronic cardiovascular use remains uncertain [40]. Pharmacological Nrf2 activators (e.g., bardoxolone methyl) robustly upregulate antioxidant responses but fail to address persistent SASP-driven inflammation and tissue remodeling

deficits.

In contrast, KA's mechanism integrates both antioxidative and anti-senescence pathways within the same therapeutic framework. By activating Nrf2/HO-1/NQO1 while suppressing the p53/p21/p16 axis, KA interrupts the ROS-DNA damage-senescence feedback loop, promotes nitric oxide bioavailability, and enhances fibrous cap stability. KA's efficacy in our model (e.g. plaque reduction and increased eNOS) was comparable to high-dose atorvastatin, yet KA also suppressed the SASP (IL-6, MMP-9), suggesting an added anti-aging benefit. This dual-action approach offers a broader cardioprotective spectrum than monotherapies, targeting oxidative injury, vascular aging, and structural integrity simultaneously. Such multi-target synergy positions KA as a promising candidate for precision cardiovascular therapy, especially in patients with metabolic syndrome or persistent risk despite optimal lipid control [38–40].

4.4. Limitations and future directions

First, we acknowledge that the optimal dosing of KA in humans remains to be established. Using body surface area scaling, our 20 mg/kg mouse dose is roughly equivalent to 100 mg/day for a 60 kg adult. This is within achievable supplement ranges: for comparison, clinical trials of related flavonoids (e.g. quercetin) have used 500 mg/day with observed cardiovascular effects [41]. We will insert these calculations and note that KA is present in many foods (e.g. kale, tea, onions). However, KA's oral bioavailability is quite low and it is rapidly conjugated in vivo [41]. For example, kaempferol glycosides (found in onions) have higher up-take than the free aglycone [41].

Translation of KA's dual-targeting, bidirectional mechanism into clinical practice will require advanced modeling and safety assessment. Organ-on-a-chip platforms have emerged as powerful tools to recapitulate human plaque microenvironments and evaluate vascular responses ex vivo [42,43]. Likewise, spatial transcriptomics offers high-resolution mapping of cell-type-specific drug effects within

atherosclerotic lesions [44,45]. Comprehensive pharmacokinetic profiling in animal models reveals that KA undergoes extensive phase II metabolism with moderate bioavailability [46,47], while human Pharmacokinetics studies confirm rapid absorption and conjugative clearance [48]. Furthermore, systematic reviews underscore the therapeutic potential of flavonoid supplementation to modulate metabolic syndrome parameters and cardiovascular risk with minimal adverse effects [49,50]. Taken together, these approaches will guide dosing optimization, safety evaluation, and personalized application of KA in cardiovascular therapy.

5. Conclusion

This study systematically elucidates the dual-target mechanism (Nrf2/p53) by which KA, an active compound derived from traditional medicinal plants, inhibits the progression of AS through synergistic regulation of the “redox-aging axis”. Our research offers a crucial theoretical basis for creating multidimensional therapeutic approaches utilizing natural medicines. This discovery not only validates the scientific principles underlying the “holistic regulation” approach in ethnomedicine but also establishes a novel paradigm for precision drug design leveraging natural products.

Authors' contribution

XYC and NG conceptualized and designed the study. NG supplied the materials and reagents. XYC conducted both in vivo and in vitro experiments. YJZ and WNZ contributed to the data analysis. XYC drafted the initial manuscript, which was subsequently revised by NG. The final manuscript received review and approval from all authors. The authors report no conflicts of interest.

CRedit authorship contribution statement

Xinyi Cai: Writing – original draft, Resources, Project administration, Data curation, Conceptualization. **Yajie Zhang:** Software, Project administration. **Weina Zhu:** Software, Resources, Investigation. **Ning Gu:** Writing – review & editing, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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